

Chemistry-informed deep learning model for predicting stereoselectivity and absolute configuration in asymmetric hydrogenation

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The asymmetric hydrogenation of olefins is one of the most important asymmetric transformations in molecular synthesis. While other machine learning models have successfully predicted stereoselectivity for reactions with a single prochiral site, existing models face limitations including narrow substrate–catalyst applicability, an inability to simultaneously predict stereoselectivity and absolute configurations in asymmetric hydrogenation of olefins with two prochiral sites, and a reliance on predefined descriptors. Here, to overcome these challenges, we introduce Chemistry-Informed Asymmetric Hydrogenation Network (ChemAHNet), a deep learning model based on the reaction mechanism of olefin asymmetric hydrogenation. By leveraging three structure-aware modules, ChemAHNet accurately predicts the absolute configuration of major enantiomers across diverse catalysts and substrates. It also defines the $\Delta\Delta G^\ddagger$ of asymmetric hydrogenation via catalyst–olefin interactions, enabling concurrent prediction of stereoselectivity and absolute configuration. Notably, ChemAHNet extends to other asymmetric catalytic reactions. By operating solely on simplified molecular-input line-entry system inputs, it captures atomic-level spatial and electronic interactions, offering a robust tool for target-directed molecular engineering.

Artificial intelligence (AI) models for predicting reaction outcomes have attracted substantial attention for their potential to accelerate synthesis design and optimization^{1–4}. Although expert-guided rule-based programs for reaction prediction have existed since the 1970s^{5,6}, advances in hardware and AI algorithms have recently led to a rapid proliferation of predictive models^{7–14}. AI-based reaction prediction is complex, hinging on robust model design and feature selection. Dominant approaches include graph neural networks (GNNs)^{7,9,11,12,15–17},

which encode the connectivity between atoms in chemical molecules, and transformer-based models^{18–22}, which represent reactions using a simplified molecular-input line-entry system (SMILES). These models show strong performance on large, diverse datasets but often fail to capture specific reaction mechanisms, leading to poor predictions for specific reactions. Incorporating chemical information, including stereochemical details, can substantially enhance the predictive performance of AI models¹. Simultaneously, enabling the model to

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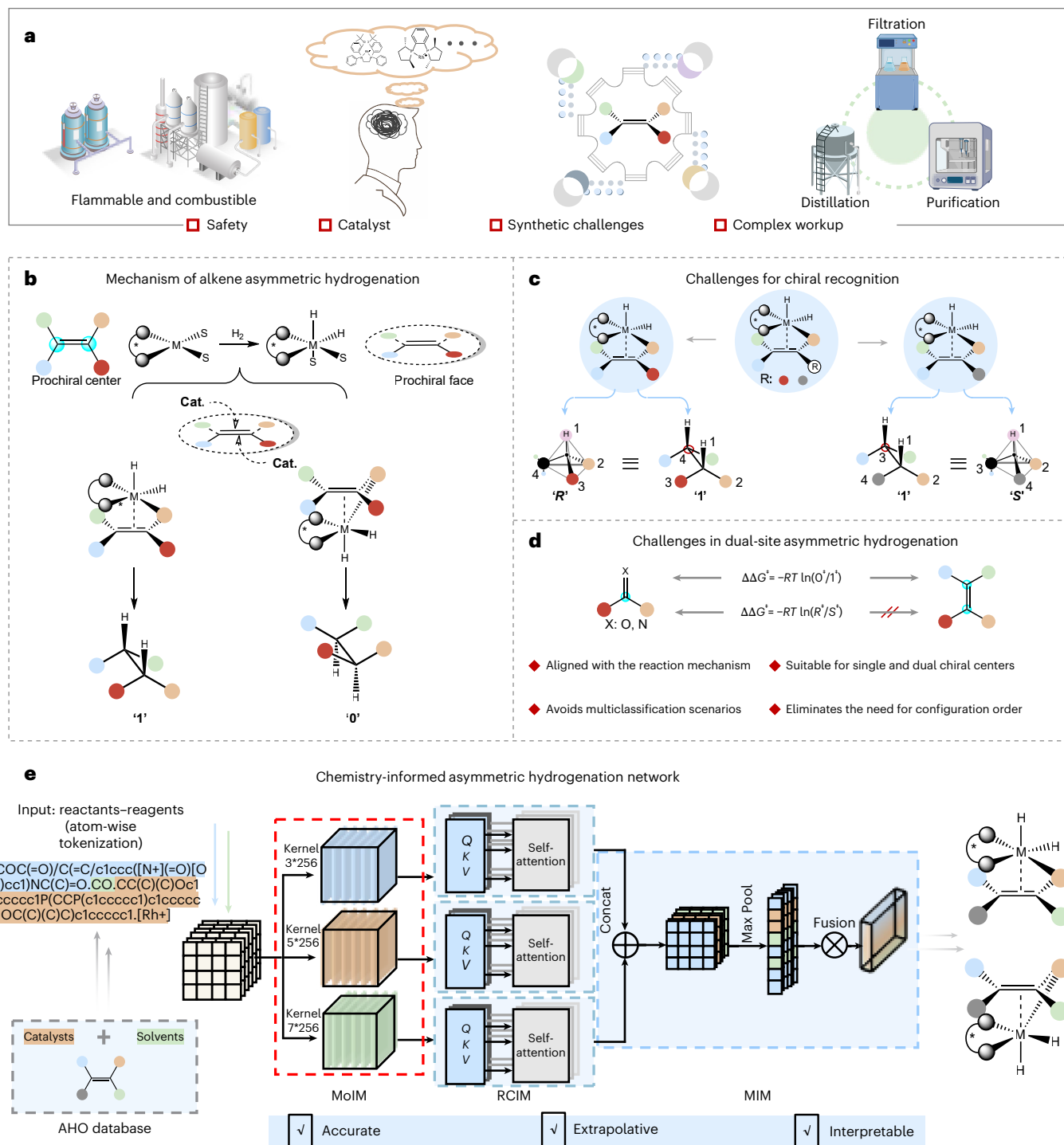


Fig. 1 | ChemAHNet: a deep learning framework for asymmetric olefin hydrogenation. **a**, Reaction complexity: asymmetric olefin hydrogenation requires the optimization of multiple conditions for efficiency and safety. **b**, Reaction mechanism and interaction mode labeling: the catalyst (cat.) interacting above the substrate is labeled '1' and below is labeled '0'. **c**, According to Cahn–Ingold–Prelog chirality rules, identical interaction modes may yield different outcomes depending on substituent priority. By directly predicting interaction modes, identical mechanisms yield consistent results, independent of group priority. The numbers 1–4 denote group priority (4 > 3 > 2 > 1). **d**, The

interaction mode enables the reasonable calculation of the $\Delta\Delta G^\ddagger$ in reactions with dual chiral centers. R is the gas constant and T is the absolute temperature (RT denotes their product). **e**, ChemAHNet workflow: input SMILES undergoes MoIM to extract moiety-level features, RCIM to integrate molecular information and MIM to model interactions and select key features for outcome prediction. The blue, orange and green shading in the ChemAHNet network conceptualize examples of different moiety combinations generated by the MoIM module. The color coding of functional groups linked to the C=C bond of the olefin represents substituents of different types and sizes covered in this study.

better capture reaction mechanisms allows for a lighter design, reducing pretraining requirements and making the model easier to train with less data and more suitable for deployment²³. When applied to emerging areas of chemistry, leveraging the essential information of

chemical reactions will enable the model to make more efficient use of valuable experimental data.

Here, we investigated an AI model that leverages reaction mechanisms to predict the products of the asymmetric hydrogenation of

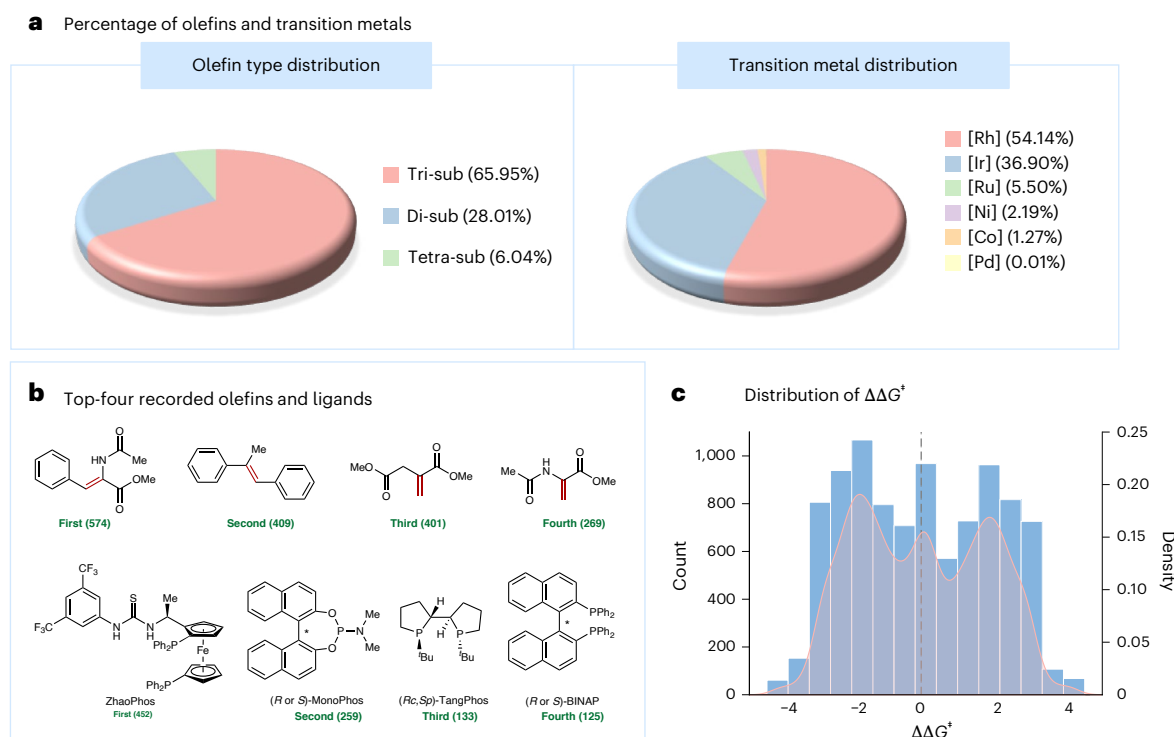


Fig. 2 | Key statistics of the created AHO database. a, The distribution of olefin substrates and transition metals. Sub, substituted. **b**, The two-dimensional structures of the top four recorded olefins and ligands. The numbers in

parentheses below each molecule represent the number of entries in the dataset. **c**, The distribution of $\Delta\Delta G^\ddagger$ values, shown as a histogram (blue, left axis: count) overlaid with a kernel density estimate (orange, right axis: density).

olefins (AHO), a task complicated by stereochemical changes and product enantiomeric ratios (Fig. 1). As a key approach in asymmetric synthesis, AHO has made a profound impact on both academic research and industrial applications²⁴. Since the Nobel Prize recognition in 2001, AHO has been established as an efficient, atom-economical strategy that reduces experimental time and lowers drug production costs^{25–27}. On the other hand, AHO requires high-pressure hydrogen, which poses safety risks. Furthermore, selecting an appropriate catalyst from a wide range of candidates and optimizing complex reaction conditions are both time consuming and labor intensive^{28–30} (Fig. 1a). Asymmetric hydrogenation of tetra-substituted olefins is particularly challenging as steric congestion and complex electronic environments hinder precise stereochemical control³¹. Therefore, an AI model that predicts the outcomes of AHO could streamline optimization and minimize unnecessary experimental trials. From a technical viewpoint, AHO embodies a challenging class of reactions for AI prediction, requiring accurate identification of substrate-to-product asymmetric transformations (absolute configuration prediction) and precise prediction of enantiomeric ratios (stereoselectivity)³².

The development of computational models for asymmetric catalysis has advanced considerably, ranging from multivariate linear regression to sophisticated machine learning algorithms^{33–39}. Despite these advancements, major challenges remain in generalizing such approaches across diverse catalytic systems. Among these, the multivariable linear regression framework pioneered by Sigman and coworkers^{33–35} has demonstrated notable success in predicting both steric effects and enantioselectivity. Subsequent studies showed that machine learning models can encode chiral information to accurately predict enantioselectivity in stereoselective transformations^{36–39}. However, contemporary machine learning implementations exhibit three fundamental limitations: First, their predictive capability is restricted to specific substrate–catalyst pairs or structurally analogous derivatives^{23,40–42}, necessitating case-specific models and impeding

broad applicability. Second, although effective for single stereocenter formations (for example, asymmetric hydrogenation of imines or ketones), these models struggle with reactions generating two chiral centers, as conventional *R/S* configurations cannot adequately distinguish such enantiomeric products. Although Hong et al.⁴³ achieved progress in predicting the stereoselectivity of alkene hydrogenation reactions yielding two stereocenters, their methodology exhibited limitations in accurately assigning the absolute configuration of the products. While their hierarchical learning strategy can enhance local predictive performance, it also increased model complexity and cost, and required retraining a dedicated model for each new substrate, constraining scalability. Third, most ML approaches depend on quantum chemical descriptors⁴⁴, which are known for their solid physical foundation and high descriptive power. However, these methods rely on complex calculations, making them slow and resource intensive for large-scale screening. To address this, we propose an AI framework that eliminates quantum chemical descriptors while predicting stereoselectivity and absolute configurations across extended substrate or catalyst libraries, enhancing olefin hydrogenation for pharmaceutical development where rapid exploration and optimization are essential.

In this work, inspired by the mechanism of olefin asymmetric hydrogenation, we designed a model framework that emphasizes the interaction mode between olefin double bonds, functional groups and catalytic metal centers (Fig. 1b). This approach reduces ambiguities in stereochemical labeling and, for reactions with dual chiral centers, helps to resolve the classification complexity of the *R/S* system and the order of configurations (Fig. 1c,d). On the basis of this concept, we developed the Chemistry-Informed Asymmetric Hydrogenation Network (ChemAHNet), which integrates three modules—moiety identification module (MoIM), reaction components integration module (RCIM) and molecular interaction module (MIM)—to capture chemically relevant features (Fig. 1e). ChemAHNet predicts both stereoselectivity and absolute configuration in olefin asymmetric hydrogenation

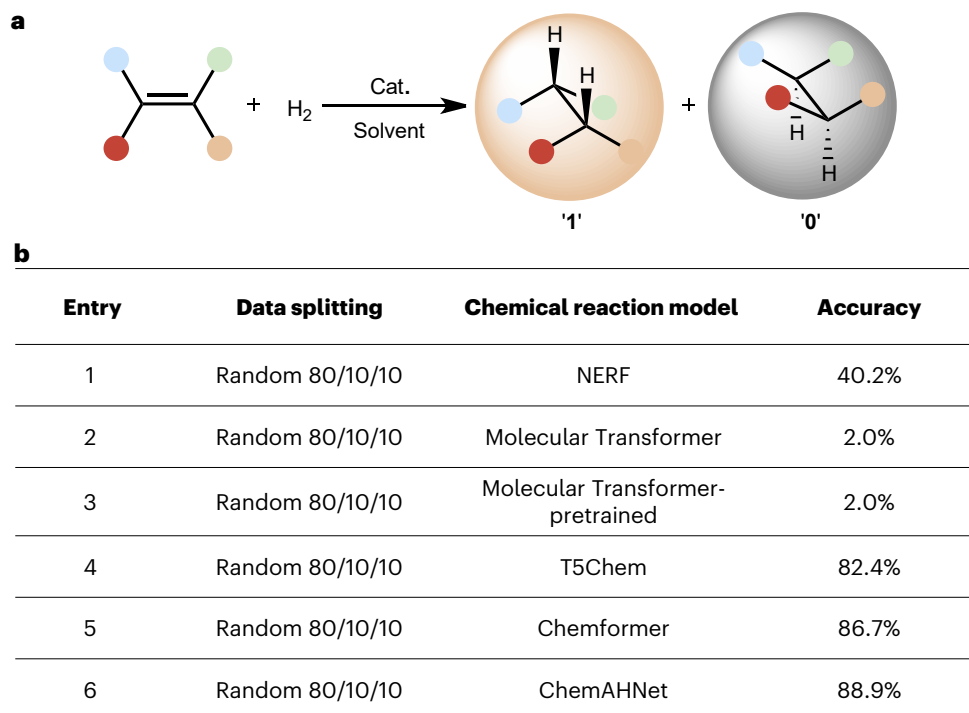


Fig. 3 | Comparison of prediction results between the ChemAHNet model and other state-of-the-art models. a, A representative reaction mode of olefin asymmetric hydrogenation. **b,** The prediction accuracy of the absolute

configuration of the major enantiomer across different models. The color coding of functional groups linked to the C=C bond of the olefin represent substituents of different types and sizes covered in this study.

with two prochiral sites without relying on chemical descriptors, and is applicable across diverse substrates and catalysts.

Results

Dataset diversity

We curated a dataset of 9,478 asymmetric hydrogenation reactions of olefins, encompassing 1,766 substrates and 1,480 catalysts across diverse conditions (Fig. 2). Substrate distribution includes tetra-substituted (6.04%), tri-substituted (65.95%) and di-substituted (28.01%) olefins (Fig. 2a). The transition metal catalysts consist of rhodium (54.14%), iridium (36.90%), ruthenium (5.50%), nickel (2.19%), cobalt (1.27%) and palladium (0.01%) (Fig. 2a). Representative olefins and ligands are shown in Fig. 2b. Enantioselectivity values $\Delta\Delta G^\ddagger$ range from -4.5 to 4.5 kcal mol $^{-1}$, exhibiting a relatively balanced distribution without skew (Fig. 2c), which supports robust model training. Overall, the dataset demonstrates a broad coverage of substrates, catalysts and enantioselectivities, highlighting its suitability for predictive modeling.

Overview of ChemAHNet architecture

We developed ChemAHNet, a chemistry-informed reaction performance model with three modules tailored for asymmetric hydrogenation. The MoIM extracts chemically meaningful substructures from substrates, ligands and catalysts. The RCIM aggregates these moiety-level features into enriched molecular representations. The MIM then models cooperative effects, particularly between olefins and catalysts, to predict stereochemical outcomes. Together, these modules enable accurate prediction of enantioselectivity and absolute configuration across diverse systems. A detailed architectural diagram, modular structure and ablation studies are provided in the Methods and Fig. 1e.

Major enantiomer prediction

Correctly predicting the experimentally observed major enantiomer is a fundamental benchmark. For AHO, we compared ChemAHNet to several state-of-the-art models (NERF¹⁵, Molecular Transformer¹⁸,

Molecular Transformer-pretrained²¹, T5Chem¹⁹ and Chemformer²⁰). The dataset was randomly split into 80% training, 10% validation and 10% testing. Detailed model replication are provided in the Supplementary Information.

Among baseline models, the graph-based NERF achieved 40.2% top-one accuracy, while Molecular Transformer and its pretrained variant performed poorly (~2%). The pretrained transformer-based models T5Chem and Chemformer reached 82.4% and 86.7%, respectively, highlighting the benefit of large-scale chemical pretraining. ChemAHNet outperformed all models with 88.9% top-one accuracy, demonstrating superior ability to capture the stereochemical outcome of olefin hydrogenation (Fig. 3).

Enantiomer prediction remains challenging because *R* or *S* assignment depends on relative substituent positioning, which may lead to opposite labels under identical mechanisms. By instead predicting the substrate–catalyst interaction mode and then inferring configuration, ChemAHNet ensures consistent outputs from the same mechanistic basis, enabling accurate recognition of stereochemical outcomes. Detailed extended discussion are provided in the Supplementary Information.

Ablation study

We assessed the contribution of each ChemAHNet module through ablation experiments (Fig. 4). Three variants were tested: (1) no MoIM, replacing the multikernel convolutional moiety extractor with a single-kernel version; (2) no RCIM, removing the RCIM; and (3) no MIM, omitting the MIM. The results (Fig. 4b) highlight the importance of all three modules. Removing MoIM reduced top-one accuracy by 3.3%, underscoring the role of multiscale functional group extraction. Removing RCIM caused a 10.1% drop, confirming that integration across reaction components is essential for comprehensive molecular representation. The most dramatic effect came from removing MIM, which reduced accuracy by 23.7%, emphasizing the critical role of explicitly modeling substrate–catalyst interactions. Together, these findings demonstrate that each module is indispensable, with MIM

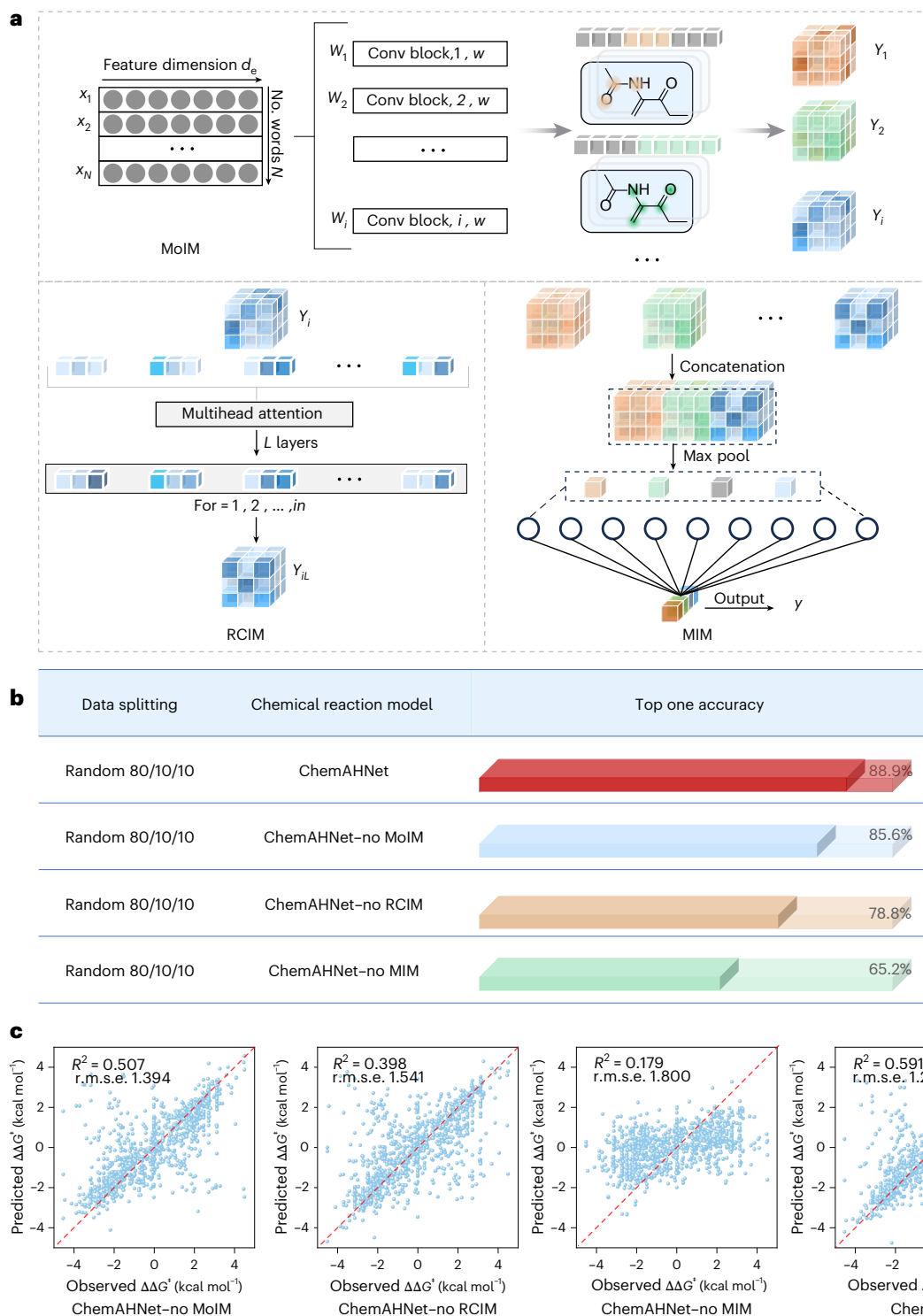


Fig. 4 | Ablation study of ChemAHNet modules. **a**, MoIM extracts moiety information from SMILES using convolutional kernels. Conv, convolution. RCIM integrates moiety data via multihead attention. MIM explores interactions using max pooling and residual networks. The orange, green and blue shading

conceptualize examples of different moiety combinations generated by the MoIM module. **b**, Enantiomer prediction showing a comparison of ChemAHNet with variants (no MoIM, no RCIM or no MIM). **c**, The regression performance showing a comparison of ChemAHNet with variants (no MoIM, no RCIM or no MIM).

being most influential for accurate enantioselectivity prediction. A detailed analysis of module design and variants is provided in the Supplementary Information.

Stereoselectivity and absolute conformation prediction

In AHO, stereoselectivity reflects the catalyst's preference for different substrate faces, determining the enantiomeric ratio (e.r.), while

absolute configuration depends on the specific interaction mode between catalyst and substrate. To unify these tasks, ChemAHNet predicts e.r. values and infers configuration directly from interaction modes (Fig. 1d), thereby overcoming inconsistencies that arise when relying solely on R/S nomenclature.

ChemAHNet reliably predicts stereoselectivity across a broad reaction space (Fig. 4c), achieving $R^2 = 0.591$ and root mean square

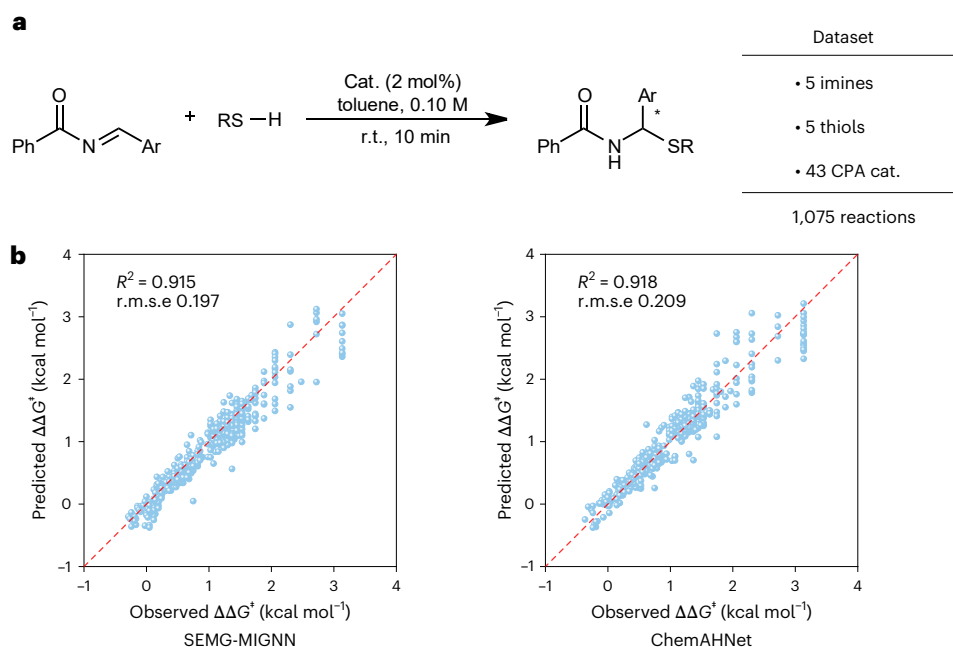


Fig. 5 | Enantioselectivity prediction for the chiral phosphoric acid-catalyzed thiol addition to *N*-acylimines using ChemAHNet and SEMG-MIGNN. **a, An overview of the dataset of the chiral phosphoric acid-catalyzed thiol addition to**

N-acylimines. These data were originally published by Denmark and coworkers. **b**, The predicted performance of SEMG-MIGNN and ChemAHNet. r.t., room temperature.

error (r.m.s.e.) of 1.27 kcal mol⁻¹. Removing MoIM, RCIM or MIM led to reduced performance ($R^2 = 0.507, 0.398$ and 0.179 , respectively), validating their design. Although the overall regression performance may not appear exceptional, ChemAHNet simultaneously predicts both stereoselectivity and absolute configuration across diverse catalysts and substrates without task-specific customization. For instance, in Rh/BINOL-phosphite-catalyzed tri-substituted olefin hydrogenations⁴³, ChemAHNet achieved $R^2 = 0.797$ and an r.m.s.e. of 0.41 kcal mol⁻¹, outperforming sterics- and electronics-embedded molecular graph with molecular interaction graph neural network (SEMG-MIGNN), differential reaction fingerprint (DRFP) and multiple fingerprint feature (MFF) baselines (Supplementary Fig. 15 and Supplementary Table 3).

We also tested the extrapolative ability of ChemAHNet. In ligand-based extrapolation, leave-one-ligand-out predictions across five scaffolds yielded R^2 values of 0.395–0.866, with classification accuracies above 85%. In metal-based extrapolation, holding out Ru, Ni and Co resulted in classification accuracies of 78–84%, although regression accuracy dropped. Finally, in value-range extrapolation, ChemAHNet maintained >84% classification accuracy for high-enantiomeric excess test sets (>97–99%), although regression performance remained moderate ($R^2 = 0.42$ –0.62) (Supplementary Information). These results indicate that ChemAHNet robustly classifies absolute configurations and provides reasonable regression predictions even under extrapolative conditions, highlighting its potential to accelerate catalyst development and asymmetric hydrogenation optimization. Detailed analyses are provided in the Supplementary Information.

Cross-domain performance validation

To evaluate cross-domain reliability, we first tested ChemAHNet on Denmark's⁴² dataset of 1,075 chiral phosphoric acid (CPA)-catalyzed thiol additions to *N*-acylimines (Fig. 5a). Following the original data split, ChemAHNet achieved $R^2 = 0.918$ and r.m.s.e. of 0.209 kcal mol⁻¹, comparable to the best-performing SEMG-MIGNN model ($R^2 = 0.915$ and r.m.s.e. of 0.197) (Fig. 5b). Across additional prediction tasks (imine, thiol, catalyst and transformation), ChemAHNet consistently yielded the best r.m.s.e. values (0.207, 0.281, 0.286 and 0.197 kcal mol⁻¹,

respectively; Supplementary Information). Unlike SEMG-MIGNN, ChemAHNet does not require quantum chemical descriptors, relying solely on SMILES-based molecular information, which improves usability and scalability.

We further assessed ChemAHNet on three curated external datasets⁴⁵ covering distinct asymmetric catalysis classes: organocatalytic conjugate additions, photoredox-catalyzed reactions and organocatalytic enamine reactions (Supplementary Information). ChemAHNet achieved R^2 values of 0.845, 0.725 and 0.829, with r.m.s.e. values of 0.340, 0.330 and 0.385 kcal mol⁻¹, respectively. While these benchmarks do not span the full diversity of asymmetric catalysis, they highlight the capacity of ChemAHNet to generalize across mechanistically distinct domains. Together, these results indicate that ChemAHNet achieves state-of-the-art performance on established benchmarks and demonstrates promising transferability beyond metal-catalyzed hydrogenation. Detailed performance metrics and additional analyses are provided in the Supplementary Information.

Model interpretation

Beyond predictive accuracy, a key advantage of SMILES-based representations is the ability to interpret model outputs at the atomic level. Since ChemAHNet relies solely on SMILES strings, predictions can be decomposed into atom-level contributions without additional quantum descriptors. To achieve this, we applied SHapley Additive exPlanations (SHAP)^{46,47} to quantify atom-wise importance scores and XSMILES⁴⁸ to visualize them (Fig. 6).

Case studies highlight that ChemAHNet identifies chemically meaningful features. For instance, in olefin substrates OL-1 to OL-3, the model correctly emphasizes the electron-donating ester and amino groups, consistent with experimental evidence that amide oxygen interacts first with rhodium in dehydroamino acid ester hydrogenations^{49,50} (Fig. 6a). In OL-4, ChemAHNet highlights the six-membered ring region under different catalysts, reflecting steric effects that align with known influences of substrate rings and ArcPhos ligand isopropyl groups⁵¹ (Fig. 6b). Finally, solvent comparisons show minimal impact on enantioselectivity (Fig. 6c), consistent with reported observations.

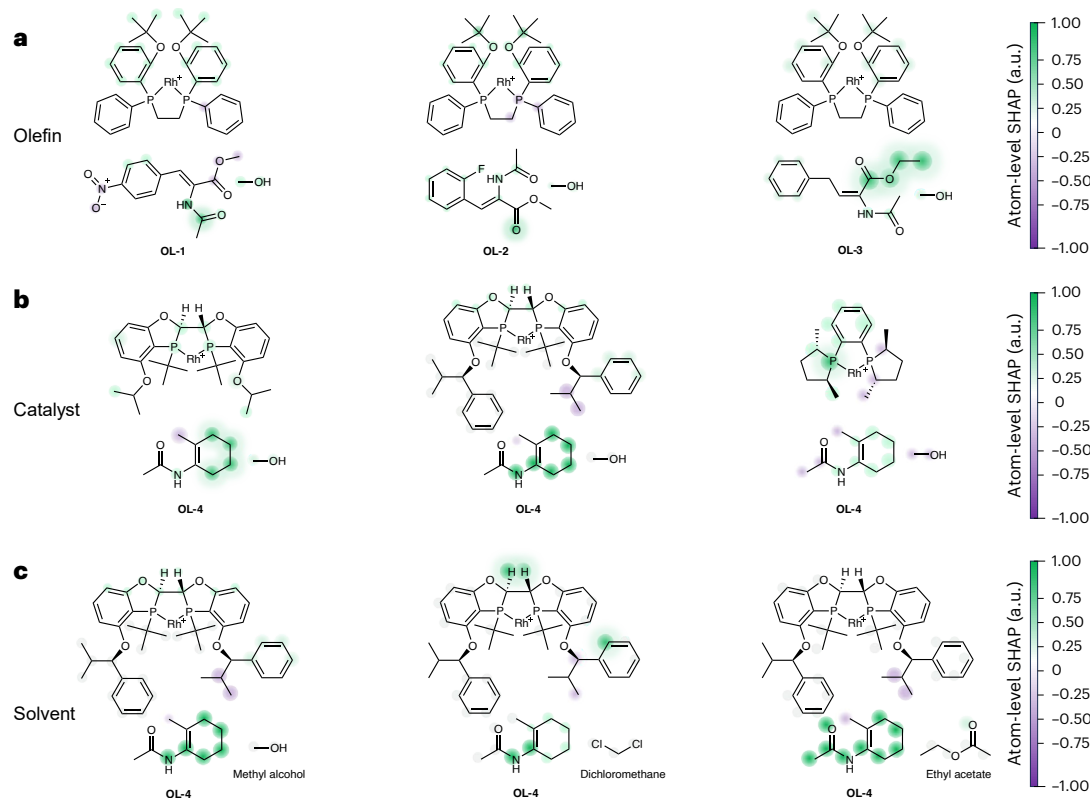


Fig. 6 | Chemical interpretation of absolute conformation and enantioselectivity determination by the ChemAHNet model. a, The chemical interpretability of ChemAHNet under different substrate conditions. **b,** The chemical interpretability of ChemAHNet under different catalyst conditions.

c, The chemical interpretability of ChemAHNet under different solvent conditions. The green shading marks moieties with positive contributions to the observed enantioselectivity and the purple with negative contributions. The numbers indicate influence magnitude.

Overall, ChemAHNet reveals electronic, steric and solvent influences that agree with mechanistic organic chemistry, underscoring its potential to guide rational catalyst and reaction design. Detailed interpretations and additional examples are provided in the Supplementary Information.

Discussion

ChemAHNet represents an important step toward integrating chemical knowledge with deep learning for asymmetric catalysis. By leveraging SMILES representations and learning atom- and moiety-level features, the model achieves reliable predictions of stereoselectivity and absolute configuration across a wide range of olefin hydrogenation reactions and demonstrates promising transferability to other asymmetric catalytic systems. Beyond predictive accuracy, the model also offers interpretability through atom-level attribution, enabling insights into electronic and steric effects relevant for catalyst and substrate design.

Despite these advances, several limitations must be acknowledged. First, the training dataset, although diverse, remains biased toward well-studied substrates, catalysts and high enantioselectivity values reported in the literature. This limits the model's performance on underrepresented functional groups and rare scaffolds, such as sulfonyl, phosphonate or silyl substituents. Second, while ChemAHNet successfully leverages SMILES strings, the absence of explicit three-dimensional structural information constrains its ability to capture conformational flexibility and noncovalent interactions, which are also important in asymmetric catalysis. Third, although SHAP-based interpretation highlights chemically meaningful regions, it cannot unambiguously distinguish between steric and electronic contributions, and therefore requires expert interpretation to link model outputs with mechanistic principles. Finally, generalization across mechanistically distinct catalytic domains, while promising,

is still preliminary and requires validation with larger, systematically curated external datasets.

Future studies could address these limitations in several ways. Expanding and balancing training datasets with underrepresented reaction classes and functional groups would improve model robustness. Incorporating three-dimensional molecular descriptors, graph-based embeddings or quantum chemistry-inspired features could better capture stereoelectronic interactions and enhance regression performance. Integrating reaction conditions such as temperature, solvent polarity and pressure would also enable more realistic predictive applications. Experimental validation of ChemAHNet's predictions, particularly in challenging systems with novel ligands or metals, will be essential to confirm its utility in guiding reaction development. Expanding cross-domain benchmarks to organocatalysis, enzymatic catalysis and photoredox systems will test the generalizability of this approach.

Overall, ChemAHNet highlights how the integration of machine learning with digitized chemical knowledge can accelerate asymmetric catalysis research, while future work should focus on broadening its applicability, addressing current limitations and validating predictions in diverse experimental contexts.

Methods

Data cleaning and labeling

The experimentally reported dataset of olefin asymmetric hydrogenation reactions, compiled by Hong Xing and collaborators, encompasses standardized information on 12,619 transformations conducted between 2000 and 2020⁴³. This dataset includes details on 2,754 olefins and 1,686 chiral transition metal catalysts. These datasets provide key statistics and a useful resource for asymmetric hydrogenation studies. However, initial quality assessment revealed substantial data heterogeneity requiring systematic recleaning, including inconsistent stereochemical

annotations and mismatched product–substrate pairings. The sequence of filtration steps involved (1) standardizing SMILES of substances (reactants, products, catalysts and solvents) through RDKit⁵², (2) removing reactions with identical reactants and products from the database, (3) eliminating records with erroneous or incomplete product stereochemical annotations and (4) deleting reaction records of *trans*-addition of alkenes from the dataset (since *trans*-addition is particularly rare in the asymmetric hydrogenation of alkenes, this reaction type was excluded from the current study²⁶). A filtering analysis, as described in the Supplementary Information, produced a curated dataset of 9,478 unique reactions after duplicate entries were removed.

In olefin asymmetric hydrogenation, the substrate interacts with the catalyst before hydrogenation. To enable stereoselectivity prediction based on substrate–catalyst interaction modes, we relabeled the dataset (Fig. 1b) as follows: (1) generating two-dimensional structures of reactants and products using RDKit, (2) comparing product and reactant conformations at the chiral site to determine the interaction mode, (3) assigning label ‘1’ if above the plane, otherwise ‘0’, and (4) converting $\Delta\Delta G^\ddagger$ values of label ‘0’ to negative, leaving those of label ‘1’ unchanged.

MoIM

At the embedding layer, each SMILES character is represented by a d_e -dimensional pretrained embeddings from ChemBERTa⁵³. These embeddings are concatenated into the matrix $X = (x_1, x_2, \dots, x_N)$, where N is the SMILES length and each x_i denotes the embedding of the i th SMILES character. To reflect the mechanism of olefin asymmetric hydrogenation, where moieties around the carbon–carbon double bond are critical, we apply multikernel convolution^{54,55} to capture moiety information of varying sizes (Fig. 4a). Adjacent embeddings are combined using a convolutional filter $W_i \in \mathbb{R}^{f \times d_e \times d_m}$, where f is the filter width, d_e is the embedding size and d_m is the output size: the output Y_i of the convolution operation for i th convolutional filter W_i can be expressed as

$$Y_i = W_i * X + \mathbf{b}_i, \quad (1)$$

where W_i denotes the i th convolutional filter, $Y_i \in \mathbb{R}^{N \times d_m}$ is the corresponding feature map, $*$ is the convolution operator and $\mathbf{b}_i \in \mathbb{R}^{d_m}$ is the bias. To better capture moiety information, we apply layer normalization to each sample. For an input vector $y_i = (y_{i1}, y_{i2}, \dots, y_{iN})$ (with length N), the layer normalization is given by

$$\hat{y}_{ij} = \frac{y_{ij} - \mu}{\sigma + \varepsilon} \gamma + \beta, \quad (2)$$

where μ and σ denote the mean and s.d. of the input y_{ij} , ε is a small constant used to avoid division by zero, and γ and β are learnable scaling and bias terms. The normalized output is $\hat{y}_{ij}$, and the final output matrix is denoted as $Y_{\text{MoIM}} = (Y_1, Y_2, \dots, Y_n)$ where n represents the number of convolutional kernels.

RCIM

After MoIM, the molecular representation is represented by $Y_i \in \mathbb{R}^{N \times d_m}$, integrating information from individual moieties and their interactions. To highlight important features and dependencies, we apply a multihead attention^{56–58} layer (Fig. 4a), composed of parallel scaled-dot attention modules whose outputs are concatenated. Each module processes three inputs: the keys (K), values (V) and queries (Q)⁵⁶, and computes attention as

$$\text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_m}}\right)V. \quad (3)$$

The dot product between the queries and keys determines their alignment, with larger values indicating closer relevance. Each key is

associated with a value vector weighted by the Softmax-normalized scores, emphasizing the most influential components. The scaling factor depends on the layer size d_m . As multihead attention lacks recurrence, we incorporate positional encodings⁵⁹, which add position-dependent trigonometric signals (equation (4)) to token embeddings of size d_m , enabling the network to capture the sequential order of functional groups

$$\mathbf{PE}_{(\text{pos}, 2i)} = \sin\left(\frac{\text{pos}}{10,000^{2i/d_m}}\right), \mathbf{PE}_{(\text{pos}, 2i+1)} = \cos\left(\frac{\text{pos}}{10,000^{2i/d_m}}\right). \quad (4)$$

Here, pos denotes the position of the functional group, $\sin$ and $\cos$ introduce position-dependent trigonometric signals, and $\mathbf{PE}_{(\text{pos}, 2i)}$ represents the vector encoding of the functional group position at dimension $2i$. In a multilayer attention network, the input to each layer is typically the output of the previous layer. After the input Y_i passes through the multihead attention layer of layer L , the final output can be expressed as

$$Y_{iL} = \text{layer}_L(\text{layer}_{L-1}(\dots \text{layer}_2(\text{layer}_1(Y_i))\dots)) \quad (5)$$

Each layer involves linear transformations of the query, key and value, attention weight computation and weighted summation. The final output $Y_{iL} \in \mathbb{R}^{N \times d_m}$ represents the molecular feature encoding after multiple attention layers, integrating functional group information. The final output matrix is denoted as $Y_{\text{RCIM}} = (Y_{1L}, Y_{2L}, \dots, Y_{nL})$.

MIM

After RCIM, the molecular representation $Y_{iL} \in \mathbb{R}^{N \times d_m}$ integrates moiety information. Since reactions rely on specific moieties, we capture interactions among reagents and select the most relevant features to predict substrate–catalyst interactions (Fig. 4a). Multiple feature maps $Y_{1L}, Y_{2L}, \dots, Y_{nL}$ are used as input; each has dimensions (N, d_i) , where d_i denotes the feature dimension of the i th input. Concatenating the input tensors along the feature dimension yields $Y \in \mathbb{R}^{N \times \sum_{i=1}^n d_i}$.

$$Y = \text{concat}(Y_{1L}, Y_{2L}, \dots, Y_{nL}) \quad (6)$$

Next, max pooling⁶⁰ with window size n is applied to Y across each feature dimension, retaining the most relevant functional group features. The pooling process can be represented as

$$\mathbf{Y}_{\text{pooled}} = \text{MaxPool}(Y, n), \quad (7)$$

which denotes the pooled output with dimension d_{pooled} . It is passed through a fully connected or convolutional layer to yield $\mathbf{Y}_{\text{transformed}}$, which is then combined with the input via a residual connection to enhance reagent interactions. After the residual connection, the output is

$$\mathbf{Y}_{\text{residual}} = \mathbf{Y}_{\text{pooled}} + \mathbf{Y}_{\text{transformed}}. \quad (8)$$

To ensure dimensional consistency, linear transformations or shortcuts are applied before residual connections^{61,62}. The resulting $\mathbf{Y}_{\text{residual}}$ passes through a fully connected layer for final predictions: $\Delta\Delta G^\ddagger$ values by regression and interaction mode classification via Softmax, yielding $\mathbf{y}_{\text{class}} \in \mathbb{R}^2$.

$$\mathbf{y}_{\text{class}} = \text{Softmax}(\mathbf{Y}_{\text{residual}} W_{\text{class}} + \mathbf{b}_{\text{class}}). \quad (9)$$

Here, $W_{\text{class}} \in \mathbb{R}^{d_{\text{residual}} \times 2}$ and $\mathbf{b}_{\text{class}} \in \mathbb{R}^2$ parameterize the classification layer. For $\Delta\Delta G^\ddagger$ numerical predictions, the output is a real number. The output $y_{\Delta\Delta G^\ddagger} \in \mathbb{R}$ represents the predicted $\Delta\Delta G^\ddagger$ value

$$y_{\Delta\Delta G^\ddagger} = \mathbf{Y}_{\text{residual}} \mathbf{W}_{\Delta\Delta G^\ddagger} + b_{\Delta\Delta G^\ddagger}, \quad (10)$$

where $\mathbf{W}_{\Delta\Delta G^\ddagger} \in \mathbb{R}^{d_{\text{residual}} \times 1}$ is the weight of the regression layer and $b_{\Delta\Delta G^\ddagger} \in \mathbb{R}$ is the bias term.

Data availability

All datasets used in this study are publicly accessible. The original AHO dataset, including the Rh/BINOL-phosphite-catalyzed hydrogenation of tri-substituted olefins, was obtained from Hong et al.⁴³ available via GitHub at <https://github.com/licheng-xu-echo/AHO.git>. The datasets curated and analyzed in the current study, including CPA-catalyzed thiol addition to *N*-acylimine, organocatalytic conjugate addition reactions, photoredox-catalyzed asymmetric reactions and organocatalytic enamine reactions, are available via Zenodo at <https://doi.org/10.5281/zenodo.17346605> (ref. 45). Source data are provided with this paper.

Code availability

All codes needed to run this model are available via GitHub at <https://github.com/CHENGLI-96/ChemAHNet/releases/tag/ChemAHNet-V1.0> and via Zenodo at <https://doi.org/10.5281/zenodo.17346605> (ref. 45).

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Author contributions

B.Z., L.C., P.-L.S. and G.X. conceived the project and designed the machine learning framework. L.C. performed the machine learning training. J. Lv, H.X. and Y.S. performed training of baseline models. J.Y., Z.X. and M.L. performed the experimental verifications. G.W., S.Z., J. Li, Z.J. and X.T. contributed to data curation. All authors are involved in the discussions and paper writing.

Competing interests

The authors declare no competing interests.

Additional information

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